

USE OF COMPUTER GRAPHICS IN DATA MANAGEMENT SYSTEMS
FOR DISTRIBUTION NETWORK PLANNING IN "ELECTRICITE DE FRANCE" (E.D.F.)

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ABSTRACT

Two data management systems for distribution network planning (L.V. and M.V. systems) are used by many EDF Distribution Centers.

The main results of these systems, the network calculations, are valorized by graphical outputs that allow :

- on the one hand, the results of the calculations to be understood and interpreted more quickly,
- and on the other, the conclusions of the studies to be conserved and communicated more easily.

In this paper are described :

- i- the data management systems for distribution network planning in use at the E.D.F.,
- ii- the management of graphical data in these systems,
- iii- the graphic input/output devices used.

INTRODUCTION

Each of the one hundred ELECTRICITE DE FRANCE (EDF) Distribution Centers is responsible for a network consisting of, on average ; 220 000 L.V. consumers, 4 300 M.V./L.V. Public Distribution substations 5 000 km of M.V. network and 5 000 km of L.V. network !

The Centers have at their disposal two distribution network management data-processing systems for the preparation of network planning and operating decisions.

Each of these systems (one reserved for low voltage , the other for medium voltage) consists of :

- a network data base describing the networks and the loads;
- a set of programs permitting network calculations or production of statistical information using this data base.

These systems are at the present time implemented on central computers connected to the Distribution Centers by consoles.

Object of the graphic representations

It was obvious even from the conception of these systems that if the results of the network calculations were to be made available to all the potential users, these results would have to be easy to assimilate and communicate. This is why programs have been developed using

suitable graphic terminals which enable maps to be drawn automatically on which network and load data taken from the data bases and the calculation results are shown simultaneously.

These maps are used by :

- the engineer responsible for the operation of the M.V. network in order to know the load flows and voltage drops over several operating schemas of the network ;
- the operating engineer whose task is to prepare proposals for the strengthening of the L.V. network ;
- those responsible for a commercial branch in order to quickly decide if a new customer can be supplied or not.

The first graphic outputs using an L.V. data base were produced on a plotting table at the EDF's Studies and Research Center in 1973.

The first network conversational computer programs were written for a graphic console in 1979. Between these two dates about thirty Distribution Centers had plotted several thousand L.V. maps and several hundred M.V. maps.

• Presentation of the present paper

The aim of this paper is to present the software and hardware choices made by the EDF with regard to the use of graphic information for distribution network management.

- Firstly, the L.V. and M.V. network management systems are described.
- Secondly, map digitizing and the introduction of graphic data into the network data bases are dealt with.
- Thirdly what the system user can have represented by the mechanical plotters is defined.
- And, finally, new uses of graphic data that can be imagined using a graphic C.R.T. are shown.

I - DISTRIBUTION NETWORK MANAGEMENT SYSTEMS

- The Low Voltage system was mainly built to use L.V. customer consumption data in order to calculate the constraints of load flows and voltage drops which take place on the M.V./L.V. networks and transformers. It is above all a load knowledge tool which is gradually replacing systematic measurements. The credibility of such a tool resides on the quality of the estimation of the power reached in network sections using knowledge of customer consumptions. 80 Distribution Centers use this system introduced in 1974 (1).

Some 50% of the French L.V. network has been recorded.

- For its part the Medium Voltage system is above all a network calculation tool. For the M.V. network the problem in fact is not one of knowing the loads since many measurements of power at the M.V. outgoing-feeder are available, but rather to calculate the constraints exerted in the network in a large number of actual or future operating situations using these loads.

The interest of the system is thus in the ease of construction of a study file and in the quality of the calculation programs connected to the data base. 40 Distribution Centers use this system which entered service in 1978. And some 25 % of the French M.V. network has been recorded (2).

The following section briefly describes :

- the contents of the network data bases,
- the structure of the data bases,
- the use of the network management systems.

1.1 - Contents of the data bases

In the data bases all the networks included between the H.V./M.V. busbars of the in-feeding primary substations and the L.V. customers are described, that is :

- at Medium Voltage : the primary substations, the M.V. lines and cables, the Public Distribution substations, the customer substations, the M.V./L.V. transformers, the schematic operational representation ;
- at Low Voltage : the L.V. lines and cables, the L.V. customers and their connections to the network.

The networks are divided up in the following way :

- at L.V. the base elements are the substation, the interrupting devices, the network sections and the customers ;
- at M.V. the base elements are the nodes (substations, interrupting devices, star points, etc) consisting of components (location of transformers, interrupting devices, etc) and the segments which connect two nodes and which consist of sections.

All these base elements are given a code of 10 or 12 characters :

- national code of the district where is situated..... : 5 characters,
- identification of the type of network element : letter 'P' (substation), 'J' (interrupting device) for a node..... : 1 character.
- number of the network element in the district..... : 4 characters,
- identification of the component of a node (if necessary)..... : 2 characters,

1.2 - Data base structure

The data are grouped in inter-related records, following a tree-like structure (the specifications of the I.M.S. language have been used).

In the conception of this structure, an electricity logic rather than a geographic logic was used so that the elements necessary to the network calculations can be easily selected and extracted. Thus the graphic information is not an "entry-point" for the system, it is only a characteristic of the network.

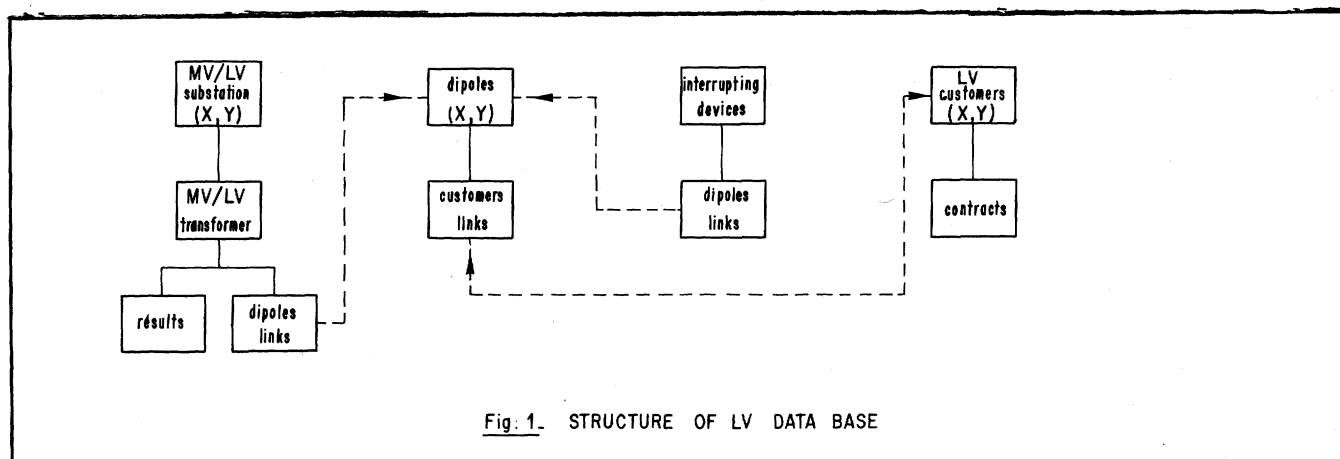
The L.V. base comprises four "physical" bases (see fig. 1) :

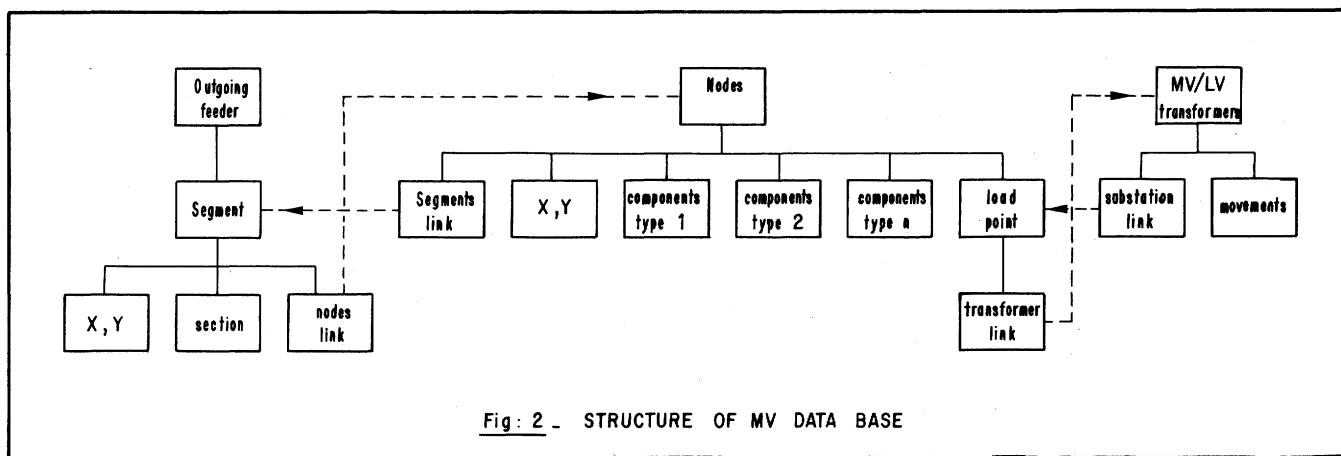
- a "substations" base describing the M.V./L.V. substations in a greatly simplified way ;
- a network sections base, in which the L.V. network characteristics are described ;
- an "interrupting devices" base ;
- an "L.V. customers" base containing past L.V. customer consumption.

The M.V. base comprises three main "physical" bases (see fig. 2) :

- a "nodes" base in which nodes and components are described ;
- an "outgoing-feeders" base describing the segments and sections grouped per "outgoing-feeder" in a schematic operational representation ;
- a "transformer" base in which the characteristics of the materials used are described.

It is estimated that for an average Distribution Center, there are some 10 000 000 characters that have to be recorded for the M.V. (5 % of which are graphic data) and 40 000 000 characters for the L.V. (25 % of which are graphic data).





1.3 - Use of the systems

. Load management and network calculations

The first use of the system is in the calculation of load-flows, voltage drops, losses in the simulated situations constructed using networks and loads contained in the L.V. or M.V. data bases.

At M.V. more complex calculation functions enable schematic operational representations to be sought which optimize certain criteria (for example, minimum failure rate or minimum losses). In addition, a simplification function allows the diminution of a network extract if wished by the operator (for example a reduction from 1000 nodes to 100 nodes) before carrying out the calculation.

. Statistics

The network descriptions were also defined in order to satisfy all the statistical knowledge needs of the network.

The user has two tools for the establishment of statistics at his disposal :

- specialised programs (lists of substations, lengths of lines, etc) for systematic statistics ;
- data base interrogation language for particular questions.

2 - DIGITIZING OF THE DISTRIBUTION NETWORK MAPS

The graphic treatments of the distribution network management systems are based upon the introduction into the data base of graphic information relative to the networks, an operation called network digitizing.

The next section deals with the following three questions :

- What information should be collected ? The nature and structure of the graphic data are stated ;
- Where is the information which is to be collected ? One is thus led to describe the preparation of the reference maps, and the collection back-up ;
- How is the information entered ? The material used to carry out the graphic data entry is described.

2.1 - Definition of graphic data

. Nature of the data

The graphic system is based upon two options :

- the networks are digitized but not the background layouts ;
- the graphic information plays no role in the dividing up and coding of the networks. It appears as complementary data to the physical description of the network and its entry is optional.

. Data recorded in the base

The graphic data recorded in the data base are thus co-ordinates (X,Y) of network element positions expressed in meters in an absolute reference system.

. For an L.V. network are recorded Substation or interrupting device

- 1 X,Y for the substation
- 1 X,Y for the place of the substation name label.

Networks sections

- 2 to 5 X,Y for the plotting of a network section ;

Customer

- 1 X,Y for the customer location.

. For an M.V. network are recorded

Nodes

- 1 X,Y for the node
- 1 X, Y to fix the position of a rectangle in which the calculation results will be written.

Segments

0 to 4 X,Y for the plotting of a segment. These values are stocked in the data bases as shown on figures 1 and 2.

2.2 - Preparation of maps for collection

The Distribution Centers have at their disposal several types at network maps.

- 1) 1/50 000th scale maps of the M.V. network,
- 2) 1/10 000th scale maps of the rural L.V. and M.V. networks,
- 3) 1/ 2 000th scale maps of the urban M.V. and L.V. networks,

4) 1/500th scale maps of the street network.

The 1/10 000th and 1/2 000th scales are generally used for the collection of the network co-ordinates.

In order to become reference maps for the collection, these maps have first of all to be "stylized" and then equipped with information.

"Stylization" of the maps

A plan legible to the naked eye is not necessarily digitizable because it may contain zones where there is high network density, in particular the urban substation outputs (H.V. or M.V.). It is thus necessary to slightly deform (viz to "stylize") the network in these zones.

Moreover, it is wished to use the system to study vast urban M.V. networks. It may be interesting to slightly deform the graphic maps in order to obtain representations which are both easy to read and to interpret. In that case, a schema of the network can be designed and recorded which no longer has any relationship with the geographical reality.

At M.V. it is possible to record several maps of the same network which enables a small urban zone to be represented either alone and in a detailed way (at 1/10 000th scale) or simplified and integrated into a rural zone (at 1/ 50 000th scale).

Map information

It is then necessary to put information on the maps which relate the network codes to the graphic data : substation names, nodes, network sections, etc.

Finally, two points, whose co-ordinates in the reference system are known, have to be put on the map. These two points enable the co-ordinates of any point in the network to be measured in the reference system using the co-ordinates measured on the original document.

2.3 - Entry of graphic data

If measurement of the co-ordinates using a simple slide rule is excluded, one is obliged to use a digitizing machine which automatically calculates the X,Y position of a point referred to on the map using a cursor and reconstructs the result discovered.

For the first experiments at the beginning of the 1970's a type of "wired logic table" was used which enabled the co-ordinates to be reconstructed on punched cards following a format defined by the wiring. Some twenty tables of this type were installed in 1973/1974 in Distribution Centers and are still in use today.

However, since 1979 numerization systems controlled by minicomputer have been used. This systems consist of :

- a digitizing table (1,20 m x 1,50 m),
- a moveable keyboard,
- a minicomputer (32 k) with screen, printer and floppy discs,
- a card puncher.

The programs which have been developed on these system enable :

- the collection to be guided by the machine using an operator dialogue ;
- instant verification of certain information : error in format or codes are signalled immediately ;
- the integration of the collections at graphic and electrical information : a "menu" system is in fact used to describe the electrical characteristics of the networks. A square on the "menu" to be found on a reserved area of the digitizing table only has to be picked out for the information corresponding to this square to be memorized.

A dozen E.D.F. Centers have been equipped with such systems during 1980.

For the moment the data collected using these systems are recorded on punched cards which are sent off to the computer which manage the data bases.

When each Distribution Center possesses its own all purpose computer a move direct connection will be established.

3 - GRAPHIC RECONSTRUCTION ON PAPER

The user having made the effort to introduce the co-ordinates into the network data base now has at his disposal many possible representations of the data and results drawn from the network management system.

The next section deals with the following three questions :

- 1) What can be obtained from graphic data ?
- 2) How does one obtain a map from the network management system ?
- 3) How are these maps produced ?

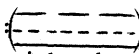
3.1 - Description of reconstructed maps (see fig. 3 and 4)

Digits not being easily understood by the human brain, part of the data from the machine is "dedigitized".

Thus in order to plot the networks :

- an X, Y becomes,
- a group of $X_j, X_j, \dots, Y_n, Y_n$ becomes 
- a code P or J becomes :  or |
- a number such as 1234 becomes : 1 2 3 4.

One of the main interests of graphic representation is the illustration of results using graphic themes. For example, the following are used :

- line segment thickness represents : (—, —, —)
 - . at L.V. : the load-flow constraints,
 - . at M.V. : the sections or capacities of the lines.
- line segment continuity represents 
 - . at L.V. : voltage drops or distinguishes between underground cables and overhead lines,
 - . at M.V. : voltage drops or the degree of line overloading or the available margin on the lines.

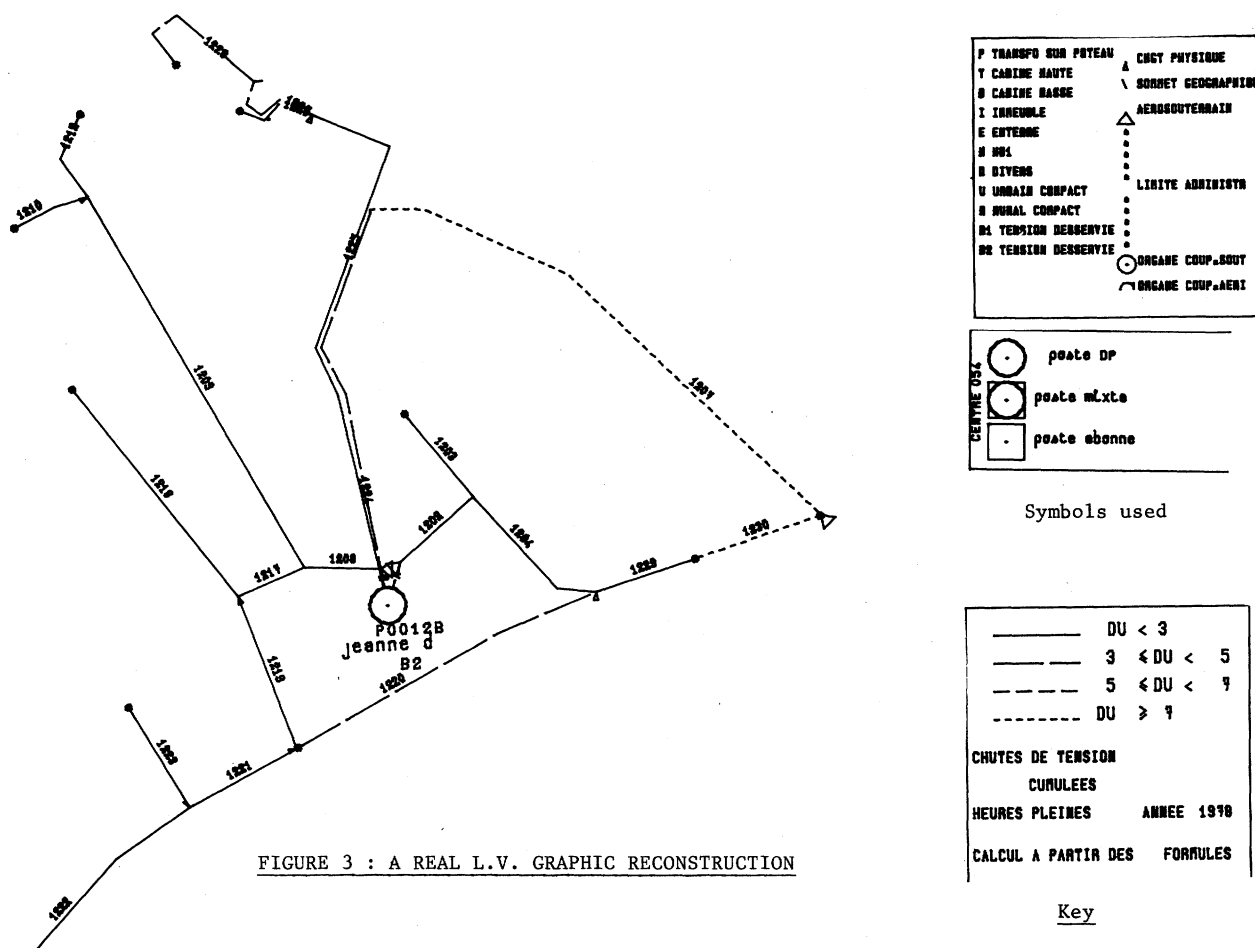
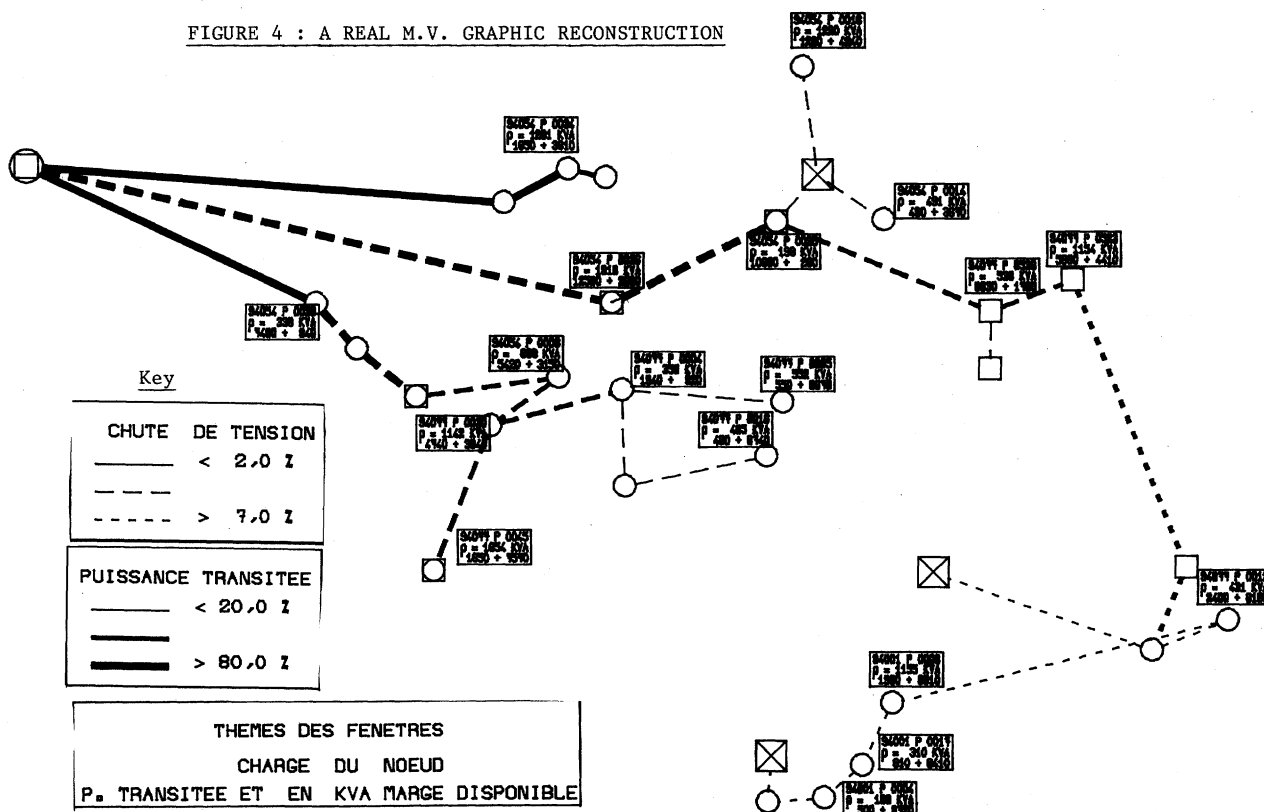


FIGURE 4 : A REAL M.V. GRAPHIC RECONSTRUCTION



- Colors are used (maximum **four**)
 - . at L.V. to distinguish the line segments and the written labels,
 - . at M.V. to distinguish the different out-going feeders in order to clearly see the operating schema.
- Some special symbols are used :

An L.V. customer's consumption can be represented by a polygon, the surface of which is proportional to the consumption.
- . There are, however, some written labels on the maps.
- At L.V. : network section numbers which are positioned automatically :
 - . inserted into the network sections for the underground networks ;
 - . parallel to the network sections for the others.
- Substation and interrupting device names, the positions of which have been numerized.
- At M.V.
 - . Written labels in the rectangle reserved for them, the position of which has been recorded. A rectangle consists of three lines of twelve characters in which can be written :
 - . first line : complete name or coded name of node,
 - . second line and third line : two pieces of information to be chosen from the following list : voltage drop, power downstream of the node, additional voltage drop for 100 kVA more, available margin, own node load, number of L.V. customers downstream of the node and power demanded by the downstream M.V. customers.

In addition, besides the area reserved for the text, the following can be written :

- a title, a legend,
- a detailed list of the physical characteristics at the L.V. network sections.

3.2 - Preparation of the maps to be edited

The plotting of a map using a plotting machine is the final step in a logical process made up of the following steps :

- extraction of the graphic and network data from the data bases,
- possible simplification of the network (at M.V. only),
- network calculation,
- preparation of graphic editing files.

- . At L.V., the user has at his disposal a control which allows him to define (3) :
 - the network to be represented while indicating the list of substations to be reconstructed,
 - the format and scale of the reconstructions ; failing which the standard format retained is 70 X 50 cm,
 - the network elements to be represented ; network sections, network section names, customer position and connection, voltage drops, etc,

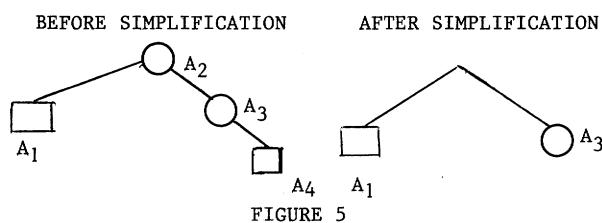
- the parameters representing the themes to be illustrated, for example voltage drop margins to be differentiated : less than 7 %, from 7 % to 15 %, more than 15 %.

The principal originality of the preparation of an M.V. map is that it includes a simplification step of the network which has been extracted from the data base.

This simplification is controlled by the user who, using a control device can state which node categories he wants to retain or dispense with. All the loads are regrouped in these nodes and the network is replaced by an equivalent network.

The following rules have been adapted for graphic simplification in order to plot automatically the simplified network (see fig. 5).

- If a final network node is removed, the segment which connects this node to the network is not represented (A4).
- If a node which connects two segments is removed, all the written labels and symbols concerning this node are removed but the adjacent segment plottings are not altered.



One is thus sure to obtain a map which is always legible. With regard to M.V. three levels of control are available for the preparation of a graphic reconstruction :

- indication of the relevant out-going feeders,
- simplification control,
- instructions concerning network calculation (type and year of calculation, network modification) and editing instruction (scales, themes to be illustrated, results to be written on the map).

3.3 - Automatic production of maps

Two types of machines are used :

- 1) Flat automatic plotting machines (100 X 120 cm, 4 colors) or roller ones (70 cm, 3 colors) which enable large format plans to be obtained, in color and of a very high graphic quality.

In addition, the flat machines enable the schema of the network to be super-imposed onto a pre-printed background map with satisfactory precision.

- 2) "Cathographe" which enables the photos cathode screen to be printed on paper 21 X 29,7 cm. The schemas thus produced are of average graphic quality and in one color only.

The main use of this machine is for the production of a full set of M.V./L.V. network substations (one picture per substation).

Communication between the computer which prepares the plotting files and the terminals which produce the maps is carried out in the following way :

- in general by magnetic tape where plotting machines operating autonomously are concerned ;
- in certain cases by liaison when the plotting machine is connected to a computer.

For the users taken as a whole, the present production of maps is the equivalent of a plotting machine operating full time.

4 - USE OF AN INTERACTIVE GRAPHIC C.R.T.

The classic map plotting system described in the preceeding paragraph produces very good quality maps, but can involve delays of several days in the reconstruction.

It thus satisfies the need to communicate and stock the data and results.

However, this system is much too "heavy" to allow the use of graphic output at all the intermediary steps of a study which may necessitate many iterations. This is why a tool has been developed able to carry out network calculations and able to show the results on a graphic C.R.T.

In the following section are described successively :

- the characteristics of the hardware used,
- the carrying out of a interactive network study,
- the communication between the minicomputer and the computer which manages the network data bases.

4.1 - Organization of the hardware components

The basic structure is an office computer combining compactly (for around \$ 25 000) :

- a 64 Kbytes minicomputer,
- a C.R.T. graphic display,
- a keyboard with programmable fonction keys,
- a thermic printer,
- a 434 Kbytes cassette memory.

This structure is experimentally installed in three Distribution Centers.

The computer language used is BASIC, extended to graphic instructions in particular.

4.2 - Carrying out a network study using a graphic terminal

The study is carried out starting from a work file extracted from the network data bases (see § 4.3).

The general plan of the realization of a study is the following :

- a) definition at the keyboard of a study hypothesis, that is, the composition using a work file of a modified load state, a modified network state, a modified operating schema state;
- b) request for the execution of load flows voltage drop and loss calculation in the hypothesis considered ;
- c) request for the graphic reconstruction on the screen of the networks and the calculation results (then go back to a)).

It is possible at any moment to request a copy of the screen, on the thermic printing device or even on a small plotting board if a better quality graphic map is required (see fig. 7).

This tool can be easily adapted to the execution of :

- all L.V. network studies,
- short term M.V. network evolution studies,
- the preparation of network control studies (seeking special operating schemas).

4.3 - Communication between the graphic console and the central computer (see fig. 6)

Present day developments aim at the following division of tasks between computers :

- data base management is carried out on a central computer. In fact the data bases have several applications and are connected to other data systems. Moreover, the management of these data bases necessitates considerable hardware and software resources,
- the interactive calculations are executed on a minicomputer using a work file taken from the data bases.

This work file is not up-dated when the network changes but it can be used for several weeks or even months to carry out studies.

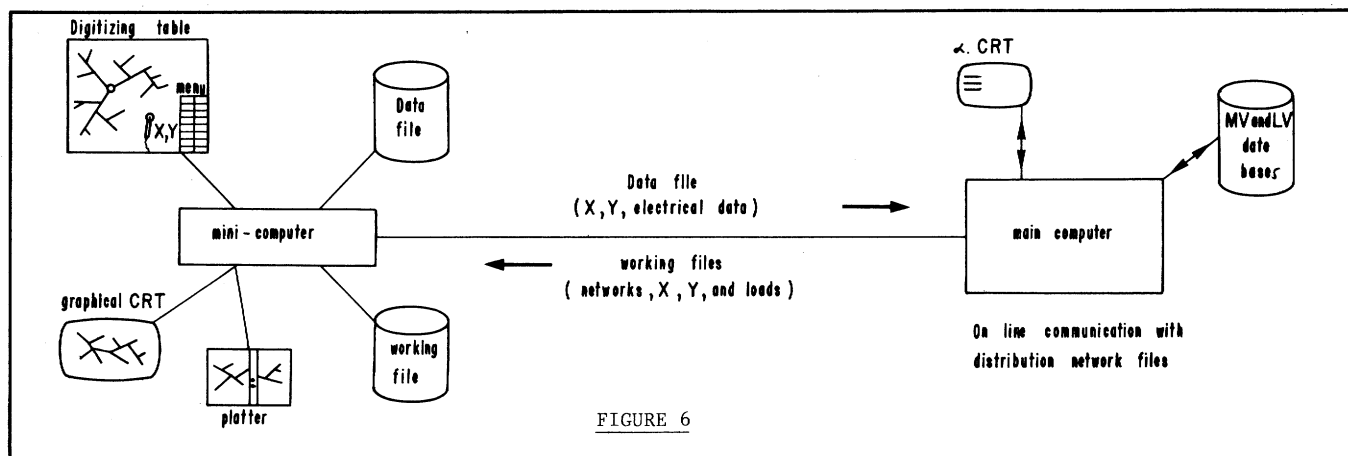


FIGURE 6

For short-term experiments ; the central computer is at one of the EDF Centralised Calculation Centers and transmission of the work file (which can reach 2 Mbytes) is done by floppy disc.

In the near future, the central computer will be the multi-purpose computer which is being installed in all the Distribution Centers and the transmission of the work file will be by direct communication.

CONCLUSION

Some ten years experience in :

- the development of graphic application programs,
- the use of specialized graphic input/output hardware
- software and hardware operation by the Distribution Centers,

have brought together a very complete set of graphic tools which satisfy the needs of network distribution management.

At the present time attention is being given to giving better user access to the tools by equipping them with a mini-computer which integrates all the specialized graphic functions as shown in fig. 6.

This integration will be carried out in laboratory tests in the coming months but the industrial realization in all the EDF Centers will take another few years.

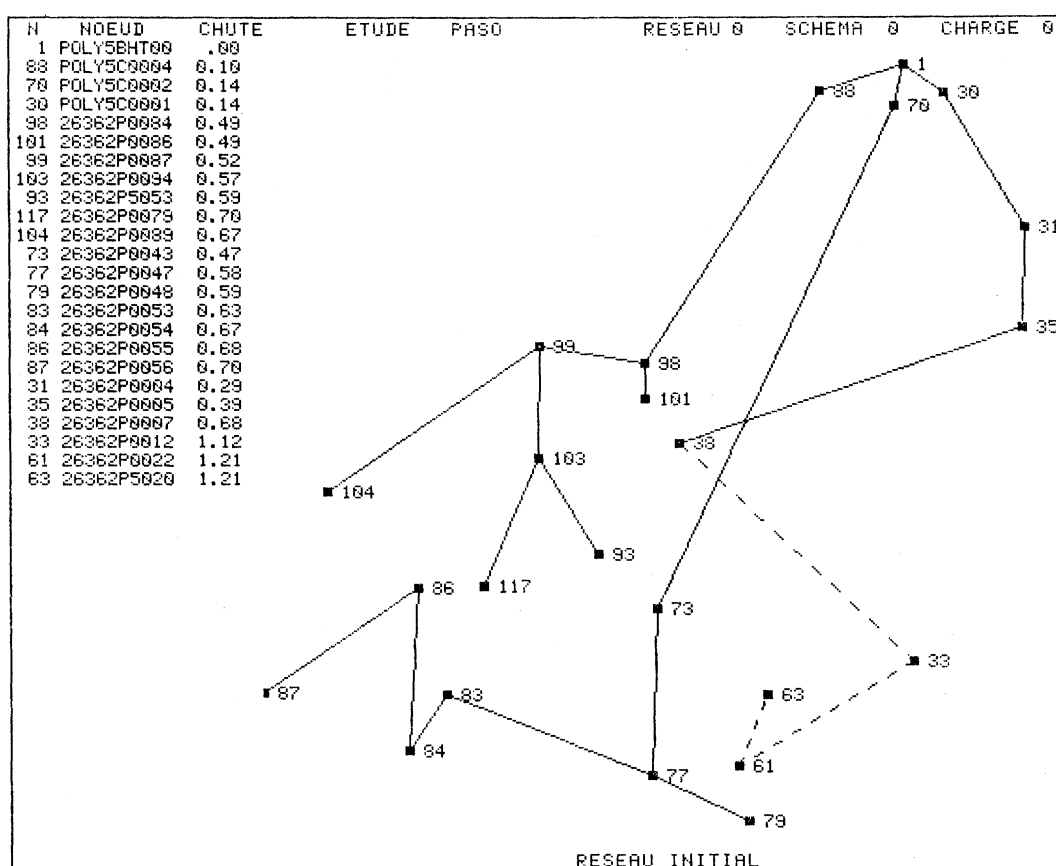


FIGURE 7 : COPY OF THE GRAPHIC C.R.T. SCREEN USING A THERMIC PRINTER

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